SGOLab: Towards Scalable Geometric Optimization

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Abstract. We present SGOLab, a research initiative that explores a geometric framework for black-box optimization. Unlike dominant paradigms that rely on stochastic sampling or heuristic reconstructions of the objective landscape, SGOLab introduces a conceptual separation between domain and codomain, incorporating geometric reference systems to guide the search. Preliminary results with the *Método de Distancias Ponderadas* (MDP) show stable performance across ranges of up to 1,000,000 dimensions, suggesting scalability beyond current approaches. More than a new algorithm, SGOLab contributes conceptual tools that enrich optimization theory, offering an open and scalable paradigm for large-scale problems.

Keywords: black-box optimization, large-scale optimization, geometric optimization

1. Introduction

Optimization in high-dimensional spaces remains a central challenge. Traditional approaches—evolutionary algorithms, gradient-free heuristics, swarm intelligence—often face scalability issues and excessive dependence on sampling.

SGOLab reframes optimization as a geometric problem, where distances and relative positions provide sufficient information to guide the search, reducing direct dependence on the codomain.

2. Innovation

The innovation of SGOLab lies in the conceptual separation between domain and codomain, the use of geometric reference systems, and the development of the prototype algorithm MDP with linear complexity $O(n)$. In addition, it introduces relative progress measures that capture activity even in scenarios where absolute improvements appear stagnant.

This approach establishes a geometric paradigm distinct from evolutionary heuristics, with potential applications in real-world problems.

3. Preliminary Results

Experiments with benchmark functions (Hypersphere, Schwefel, Ackley, CEC17) show:

- Scalability up to 1,000,000 dimensions, rarely addressed in the literature.
- Robustness in noisy and high-dimensional environments.
- Consistency across multiple test scenarios, reinforcing the exploratory nature of the framework.

The initial goal is not to prove absolute optimality, but rather stability and viability under extreme conditions.

4. Potential Impact

- **Scientific contribution:** introduces a geometric paradigm complementary to classical theory.
- **Applications:** bioinformatics, materials design, hyperparameter tuning in machine learning, among others.
- **Open science:** developed under principles of reproducibility and collaboration, fostering community validation.

5. Current Status and Future Work

Two main directions are currently being explored:

- **Hybridization with other methods:** integration with population-based or gradient-based approaches.
- **Hypothesis for accelerating the rate of convergence:** investigating how the base unit of distance and reference systems can accelerate convergence toward optimal solutions.

The greatest potential of SGOLab lies not only in the development of a new algorithm, but in the conceptual tools it contributes to optimization theory.

6. Invitation to Collaborate

SGOLab invites researchers and professionals to participate in the validation and expansion of this geometric framework. Opportunities include algorithm development, comparative evaluation against classical methods, and applications to real-world problems.

7. Conclusion

SGOLab opens a new paradigm in geometric optimization, demonstrating viability at extreme scales and offering an open framework for the scientific community. Its preliminary results highlight its potential, while its orientation toward open science ensures accessibility and collective progress.

Additional Resources

For those interested in exploring the evolution of the project and accessing complementary material:

- **Initial phase (MDP):**
  `https://www.amoaiq.com/publicaciones/#flipbook-df_990/146/`
- **Current state of SGOLab:**
  `https://github.com/misa-hdez/sgo-lab`